Preliminary Communication

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Epigenomics

Comprehensive data analysis for development of custom qRT-PCR miRNA assay for glioblastoma: a prevalidation study

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Aim: Glioblastoma (GB) is one notable example of miRNA-modulated neoplasms. Given its unique expression signature, proper miRNA profiling can help discriminate between GB and other types of brain tumors. The current work aimed to develop a more GB-specific and applicable custom designed qRT-PCR miRNA assay.**Materials & methods:** A comprehensive data analysis of bioinformatics databases, previous literature and commercially available predesigned miRNA PCR arrays within the market. **Results:** A highly enriched panel of 84 deregulated and GB-specific miRNAs has been developed. **Conclusion:** After validation of this newly developed array, it can not only save the researcher's time and effort but can also have a potential diagnostic and/or prognostic role in GB, paving the road toward personalized medicine.

Graphical abstract:

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Glioblastoma (GB) is the most common primary malignant brain tumors in adults [1]. GB incidence ranges from 4.67 to 5.73 cases per every 100,000 [2]. It is highly aggressive and graded by WHO as grade IV [1]. Despite the progress of innovative modalities in the diagnosis and treatment of such lethal cancer, the outcome for most diagnosed patients remains poor with a median survival of 15 months to 3 years [3]. Due to this sustained poor overall survival, a new array of diagnostic and prognostic indicators is requested to facilitate the clinical decision making and the individualized targeted therapy. Recent multiomics studies have revealed the heterogeneity and the complexity underlying GB at the genetic and epigenetic levels [4,5].

One of the most novel biological discoveries in the last decades is miRNAs. They are small ncRNAs (<200 nucleotide long), which act as central regulators of gene expression [6]. They are not only involved in numerous physiological, biological processes and pathological conditions, but can also play a key role as promising theranostic biomarkers [7]. According to the latest version of miRBase 22 released in March 2018 (http://www.mirbase.org/), there are 2694 annotated mature miRNA sequences in human [8].

MiRNA biogenesis involves two main steps, one in the nucleus and the other in the cytoplasm [6]. Initially, they are transcribed from their corresponding genes with RNA polymerase II/III to form a primary transcript called pri-miRNA with hairpin loop [9], which is then cleaved by Drosha/DGCR8 nuclear microprocessor complex to form a precursor miRNA (pre-miRNA) that is exported outside the nucleus by exportin-5 and Ran-GTP proteins [10]. In the cytoplasm, pre-miRNA is further trimmed to generate a miRNA duplex [11]. One of the

Figure 1. Selection of the putative candidate miRNAs in the discovery phase.
HMDD: Human miRNA and disease database.

strands is usually degraded, while the other forms a complex with RNA-induced silencing complex (RISC). Within RNA-induced silencing complex, the mature miRNA base pairs with the mRNA target at a complementary sequence driving its degradation or translational inhibition [12]. One miRNA can negatively regulate the expression of hundreds of protein-coding and noncoding genes. On the other hand, a single target can be influenced by many miRNAs simultaneously. Thus, miRNAs can form complex gene regulatory networks, which impact a wide range of cellular pathways, from development to oncogenesis [13]. This finding renders miRNAs promising novel therapeutic targets [14].

miRNAs are tightly regulated at various cellular levels under temporal and spatial control. Expression of miRNAs could be influenced by their chromatin remodeling, transcriptional and post-transcriptional regulation, dietary and hormonal factors [7]. Deregulated transcriptomic signature of miRNAs is a common feature in human diseases [15,16]. Overexpression or downregulation of miRNAs had been reported in all types of tumors and was correlated with cancer development and progression [17]. Importantly, miRNAs were found to be expressed in a tissue-specific manner [18], thereby paving the way for their utility as diagnostic and prognostic markers.

Recently, many bioinformatics tools have emerged to manage the archiving and interpretation of miRNA data in a systematic framework to explore the role of miRNAs in various diseases [13]. By using these computational web servers, functional analysis using combined experimental and predicted data, unleash targeted pathways and gene ontology (GO) of miRNAs as well as underlying transcriptional mechanisms that are hidden within complex interactions and genome-wide expression datasets [19].

Throughout the first stage of our project implementation on the miRNA expression signature profiling of GB cases in our region (middle east), we searched for the commercially available predesigned miRNAs PCR array specific for GB and we found an interassay inconsistency in their miRNA expression profile platforms. In addition, some arrays lacked specificity to the GB pathological type of brain cancers. Hence, we aimed to do a comprehensive data analyses collected from several resources, including the bioinformatics databases, previous literatures and the available predesigned miRNA PCR arrays in the market, aiming to establish a more GB-specific, cost-effective and applicable custom-designed (96-well plate) qRT-PCR miRNA transcriptomic panel. Application of this newly designed array especially in geographic regions with a limited financial support, after its validation in the second stage of our project, will save time and efforts for future researchers and clinicians in this devastating disease.

Methodology

Exploring company arrays for brain cancer

Commercially available 96-well plate arrays used in the market were screened. Comparison between miRNA panels were carried out to identify the common markers that are specific for GB.

Identifying GB-related miRNAs by bioinformatics tools

As shown in Figure 1, the selection of the candidate GB-related miRNAs included: DIANA Tools Online Suite (DIANA-microT-CDS, DIANA-TarBase v7.0, and DIANA-miRPath v3.0) (http://diana.imis.athena-innovation.gr/), which explored deregulated miRNAs involved in the pathogenesis of glioma pathway and identified putative predictive and manually curated experimentally validated targets of these miRNAs, in particular, those involved in the glioma pathogenesis (http://www.microrna.gr/tarbase). Next, functional enrichment pathway and GO analysis were carried out (http://www.microrna.gr/miRPathv2) [20]. Analysis algorithms for miRNA–gene interactions

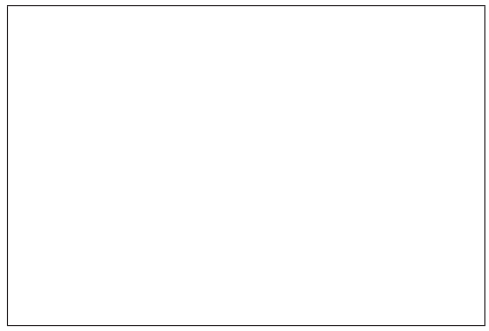

Figure 2. Three phases of biomarker panel development.

stored in DIANA relational database were run in R. Fisher's Exact test (hypergeometric distribution) and Benjamini–Hochberg's False Discovery Rate methodologies were used for statistics. Combining results options (genes union and pathways union) were selected.

Exploring miRNAs–disease interactions

Validating the enrollment of the selected miRNAs in glioma and GB was executed using several disease-associated miRNA–target interactions webservers; including miRTarBase database (http://miRTarBase.mbc.nctu.edu.tw/), which contains data validated experimentally by reporter assays, western blot or microarray experiments with overexpression or knockdown of miRNAs; human miRNA and disease database (HMDD) v2.0 (http://www.cuilab.cn/hmdd) for comprehensive annotations to the human miRNA–disease association data [21]; miR2Disease database (http://watson.compbio.iupui.edu:8080/miR2Disease/index.jsp), which contains a comprehensive resource of miRNA deregulation in various human diseases (Figure 1) [22].

Expression levels of miRNA panel in brain cancer

Literature search was done in PubMed databases and Google Scholar search engine to identify research articles and prior meta-analysis studies on 'miRNAs expression' and 'glioblastoma OR glioma' (last search update: 20 March 2018). Transcriptomic data were extracted in preformed excel sheet.

In addition, molecular cancer datasets for GB were screened from Gene Expression Omnibus [23] and The Cancer Genome Atlas [24] for miRNA-expression profiling. Integration of output was performed using the web browser of the database of differentially expressed miRNAs in human cancers version 2.0 (dbDEMC 2.0) (http://www.picb.ac.cn/dbDEMC/index.html?vhydxsbmfbtdqnrz) [25]. The Limma package embedded in R was employed by the program to screen the deregulated miRNAs collected from different experiments across multiple tumors.

Selection of endogenous controls

We surveyed the literature [26,27] and company products for the optimum reference housekeeping genes for miRNA profiling in brain tissue, aiming at finding genes that had stable expression in normal and disease states.

Development of the brain cancer miRNA profiling panel

Gene union of the aforementioned sources, filtration, duplication removal and an intersection were executed for final selection of the highly impacted GB-related miRNAs. Subsequent designing of miRNA assay location and plate layout were done to fulfill the first phase of the workflow of biomarker development (Figure 2).

Deciphering functional enrichment analysis of the selected panel

The predicted target genes of selected miRNAs were analyzed for GO terms and Kyoto encyclopedia of genes and genomes (KEGG) enrichment pathways using DIANA-miRPath v3.0 as previously described [15].

Results

Disease-focused ready-made panels for brain cancer

Four ready-made arrays were determined from the literature and companies websites (Table 1). First, a panel of 84 miRNAs in miScript™ miRNA PCR Array Human Brain Cancer (cat. no. MIHS-108Z) is currently used

Table 1. Disease-focused ready-made panels for brain cancer.

Company	Product year	Catalog number	miRNA PCR array	No. of miRNAs
Qiagen†	2011	MIHS-108Z	miScript™ miRNA PCR Array. Human Brain Cancer	84
Exiqon‡	NA	YAHS-102Y	Human Cancer Focus, miRCURY LNA miRNA Focus PCR Panel	84
GeneCopoeia§	2013	PAM-HCN96	miProfile human brain cancer miRNA qPCR arrays	84
Life Technology (Thermofisher)¶	2005	NA	NA (only customer-made array)	180

N.B., Since 1 October 2017, Exiqon products have become integrated with Qiagen company.
Data source:
† https://www.qiagen.com/us/shop/pcr/primer-sets/miscript-mirna-pcr-arrays/?catno=MIHS-108Z#geneglobe
‡ http://labomics.com/index.php/products/micrornas/miprofiletm-mirna-qpcr-arrays.html
§ http://www.genecopoeia.com/wp-content/uploads/2013/09/Gene_and_miRNA_qPCR_Arrays.pdf
¶ https://tools.thermofisher.com/content/sfs/brochures/taqman-microrna-assay-faqs.pdf and https://www.gene-quantification.de/rizdon-AB.pdf
NA: Not available; N.B.: XXX.

Table 2. Gene list of Brain Cancer miScript miRNA PCR Array panel of Qiagen company.

Type of brain cancer	Gene list	No.	% of panel
Glioma	miR-106b-5p, miR-10b-5p, miR-128, miR-130b-3p, miR-144-3p, miR-146b-5p, miR-148a-3p, miR-15a-5p, miR-15b-5p, miR-16-5p, miR-17-5p, miR-181a-5p, miR-181b-5p, miR-182-5p, miR-183-5p, miR-184, miR-19a-3p, miR-200a-3p, miR-20a-5p, miR-21-5p, miR-210, miR-221-3p, miR-222-3p, miR-23a-3p, miR-25-3p, miR-26a-5p, miR-27a-3p, miR-296-5p, miR-328, miR-34a-5p, miR-451a, miR-486-5p, miR-9-5p, miR-9-3p, miR-93-5p	**35**	**42**
Glioblastoma	miR-107, miR-10b-5p, miR-128, miR-130a-3p, miR-153, miR-16-5p, miR-181a-5p, miR-181b-5p, miR-185-5p, miR-191-5p, miR-21-5p, miR-221-3p, miR-222-3p, miR-222-5p, miR-23a-3p, miR-23b-3p, miR-25-3p, miR-34a-5p, miR-425-5p, miR-451a, miR-486-5p, miR-7-5p	**22**	**26**
Giloblastoma multiforme	miR-101-3p, miR-107, miR-10b-5p, miR-124-3p, miR-128, miR-129-5p, miR-132-3p, miR-133a, miR-133b, miR-137, miR-138-5p, miR-149-5p, miR-153, miR-185-5p, miR-187-3p, miR-203a, miR-21-5p, miR-210, miR-218-5p, miR-221-3p, miR-29b-3p, miR-31-5p, miR-323a-5p, miR-328, miR-7-5p	**25**	**30**
Anaplastic astrocytoma	miR-107, miR-10b-5p, miR-124-3p, miR-125b-5p, miR-127-5p, miR-129-5p, miR-137, miR-138-5p, miR-187-3p, miR-203a, miR-218-5p, miR-296-5p, miR-31-5p, miR-320a, miR-7-5p	**15**	**18**
Medulloblastoma	miR-106b-5p, miR-124-3p, miR-125a-5p, miR-125b-5p, miR-127-5p, miR-128, miR-129-5p, miR-130a-3p, miR-130b-3p, miR-133b, miR-137, miR-138-5p, miR-148a-3p, miR-150-5p, miR-153, miR-17-92 cluster (miR-17-5p, miR-18a-5p, miR-19a-3p, miR-20a-5p, miR-19b-3p, miR-92a-3p), miR-17-3p, miR-187-3p, miR-190a, miR-191-5p, miR-203a, miR-210, miR-216a-5p, miR-217, miR-218-5p, miR-25-3p, miR-29a-3p, miR-29b-3p, miR-30b-5p, miR-30c-5p, miR-320a, miR-323a-5p, miR-324-5p, miR-326, miR-328, miR-331-5p, miR-335-5p, miR-34a-5p, miR-425-5p, miR-7-5p, miR-9-3p, miR-93-5p, miR-96-5p	**49**	**58**
Meningioma	let-7b-5p, miR-106a-5p, miR-106b-5p, miR-125a-5p, miR-153, miR-15a-5p, miR-181a-5p, miR-185-5p, miR-19b-3p, miR-200a-3p, miR-217, miR-23b-3p, miR-26a-5p, miR-27a-3p, miR-29a-3p, miR-335-5p, miR-451a	**16**	**19**
Neuroblastoma	let-7b-5p, miR-106a-5p, miR-107, miR-125a-5p, miR-125b-5p, miR-128, miR-129-5p, miR-137, miR-141-3p, miR-146a-5p, miR-149-5p, miR-150-5p, miR-17-3p, miR-181a-5p, miR-181b-5p, miR-184, miR-187-3p, miR-190a, miR-200a-3p, miR-216a-5p, miR-221-3p, miR-23b-3p, miR-24-3p, miR-29a-3p, miR-29b-3p, miR-29c-3p, miR-30b-5p, miR-30c-5p, miR-320a, miR-323a-5p, miR-324-5p, miR-326, miR-331-5p, miR-335-5p, miR-34a-5p, miR-9-5p, miR-92a-3p, miR-93-5p	**37**	**44**
Retinoblastoma	miR-320a	**1**	**1**

https://www.qiagen.com/us/shop/pcr/primer-sets/miscript-mirna-pcr-arrays/?catno=MIHS-108Z#geneglobe

as a product in the Qiagen company to analyze mature miRNA by using SYBR® Green-based real-time PCR technology. It contains miRNAs known to be deregulated during nervous system-related carcinogenesis (Table 2). After exclusion of miRNAs, which are more related to other types of brain cancer and removal of the duplicates between glioma and GB disease, we selected 59 out of the 84 miRNAs enrolled in glioma and GB diseases for downstream analysis (Supplementary Table 1). Second, Exiqon company produces a ready-to-use Focus PCR panels of miRNA assays in a 96-well plate for all types of cancer without specification to brain cancer (Human Cancer miRCURY LNA miRNA Focus Panel, cat no. YAHS-102Y) (Supplementary Table 2). These 84 miRNAs have been shown to be important for tumorigenesis and cancer diagnostics. Third, GeneCopoeia products include miProfile miRNA PCR arrays that are designed for profiling the expressions of predefined sets of miRNAs in various tissues or cells. Human brain cancer 96-well plate (PAM-HCN96) also contains 84 pairs of PCR primers with specific forward primers and universal reverse primers (Supplementary Table 3). These miRNAs are of biological significance and importance, relevant to cancer research. Fourth, Applied Biosystems (Thermo Fisher Scientific

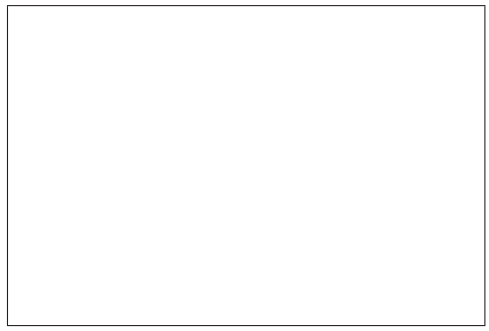

Figure 3. Comparison between four commercial ready-made miRNA panels. A total of 169 miRNAs was included from the four panels. Venn diagram discriminates the unique and common features of each panel. Circled in blue the unique group of miRNAs in each panel.

Co.) products customed a plate in 2005 (Supplementary Table 4). The intersection of the miRNAs listed in each ready-made panel is depicted in Figure 3. A total of 169 miRNAs was included from the four panels, only 10 miRNAs (5.9%) were common in all of them, while more than half of the total miRNAs were unique in each panel, accounting for 91 (53.8%) miRNAs.

Bioinformatics selection of miRNAs

We searched for miRNAs involved in glioma pathway (KEGG ID 05214) using DIANA miRPath v3.0, which revealed 587 enriched miRNAs (Supplementary Table 5). Fisher's Exact Test (Hypergeometric distribution) was adjusted as the enrichment analysis method and p-value threshold was set below 0.05. The top 99 highly significant miRNAs were selected for further downstream analysis, and their experimentally validated gene targets in glioma pathway were identified using miRTar v7.0 (Supplementary Table 6). Functional enrichment analysis of the selected miRNAs is demonstrated in Supplementary Table 7.

Exploring miRNAs & glioma/GB interactions in literature

Searching in miRTarBase, a miRNA–target interactions database validated experimentally by reporter assay, western blot, microarray or next-generation sequencing experiments, showed 84 miRNAs related to glioma and 118 miRNAs for GB (Supplementary Table 8). Human miRNA and disease database annotated a total of 204 miRNAs, including 118 miRNAs for GB and 86 for gliomas (Supplementary Table 9). Regarding miR2Disease database, 41 and 25 miRNAs were identified in GB and glioma, respectively, with ten miRNAs common between them (Supplementary Table S10).

Screening deregulated miRNAs in brain cancer

Surveying several articles illustrated deregulated miRNAs in GB; 145 upregulated and 147 downregulated miRNAs (Supplementary Tables 11 & 12). Microarray data screening by dbDEMC identified two experiments for GB in which brain cancer tissues were compared with normal brain tissues. In the first experiment (GSE13030), 93 cases and 93 controls were enrolled, and 260 miRNAs were analyzed, whereas, in the second dataset (GSE25631), 857 miRNAs were assessed in 82 cancers and 5 control tissues (Supplementary Table 13). The results of these microarray data identified 170 and 65 differentially expressed miRNAs, respectively (Figure 4A–C). The top upregulated miRNAs were miR-138, -21, -347, -196a and -221, while the lowest fold changes were for miR-18b, -181a, -15, -128a and -181c (Figure 4D). Expression levels of miRNAs (34 upregulated, 188 downregulated and 3 with inconsistent direction) are listed in Table 3 and Supplementary Table 14.

Identification of a set of reference genes for miRNA profiling

Regarding the optimum housekeeping genes for analyzing miRNAs in brain tissues, inconsistent results in literature were found. Exiqon company uses the following reference genes: SNORD38B, SNORD49A and RNU6B, while Qiagen company plate included three other small nucleolar RNAs (SNORD61, SNORD95, SNORD96A). Those six genes will be used in our future wet lab work for data normalization using the δ–δ threshold method of relative quantification [28]. *Caenorhabditis elegans* miR-39 primer assay will be used as an alternative data normalization tool.

Figure 4. Results of molecular microarray cancer datasets. (A) Comparison between two datasets on glioblastoma. Venn diagram shows common and unique deregulated miRNAs in both datasets and listed the common upregulated (red) and downregulated (green) miRNAs. **(B)** Scatter plot showing the fold change (log transformed) of differentially expressed miRNAs. Upregulated miRNAs in pink and downregulated miRNAs in green. **(C)** Pie chart showing the proportion of the direction of miRNA expression profile. **(D)** Bar chart demonstrating the top up- and downregulated miRNAs in the two experiments.

Table 3. Expression changes of miRNAs in GBM tumors in microarray datasets GSE25631 and GSE1330.

Fold change	miRNA representatives	No.	%
Up two- to eightfolds	miR-138, miR-21, miR-347, miR-196a, miR-221, miR-1251, miR-142-5p, miR-21†, miR-1537, miR-801, miR-196b, miR-155†, miR-188-3p, miR-19a, miR-214†, miR-, 9b-1†, miR-199b-5p	17	11
Up one to twofolds	miR-550†, miR-16-1†, miR-551b, miR-138, miR-373, miR-9, miR-193a-3p, miR-301a, miR-339-5p, miR-347, miR-542-3p, miR-55, miR-362-3p, miR-24-2†, miR-18a, miR-42-3p, miR-19b, miR-220c, miR-497	19	12
Down two- to eightfolds	miR-18b, miR-181a, miR-15a, miR-128a, miR-181c, miR-16, miR-876-3p, miR-1224-5p, miR-203, miR-138-2†, miR-873, miR-338-5p, miR-383, miR-7-2†, miR-139-3p, miR-522, miR-518e†, miR-770-5p, miR-1250, miR-128b, miR-298, miR-1229, miR-148b, miR-433, miR-490-5p, miR-323-5p, miR-518b, miR-153, miR-452, miR-375, miR-341, miR-421, miR-200b, miR-218, miR-338, miR-516-5p, miR-291-3p	37	24
Down one to twofolds	miR-526c, miR-668, miR-193, miR-485-5p, miR-370, miR-760, miR-351, miR-333, miR-380-3p, miR-154, miR-329, miR-352, miR-146b, miR-935, miR-138-1†, miR-9b-2, miR-339, miR-190, miR-371, miR-18, miR-7, miR-137, miR-1249, miR-207, miR-23b†, miR-132, miR-469, miR-373, miR-139-5p, miR-326, miR-185, miR-448, miR-376b, miR-628-5p, miR-208, let-7d, miR-503, miR-485-3p, miR-212, miR-217, miR-299-5p, miR-322, miR-30c, miR-539, miR-199a, miR-300, miR-344, let-7e, miR-199b, miR-432, miR-194, miR-291-5p, miR-495, miR-188, miR-328, miR-517a, miR-523, miR-150, miR-19b-1, miR-294, miR-410, miR-222, miR-34b, miR-33, miR-519d, miR-323, miR-34c, miR-181b, miR-20, miR-515-3p, miR-431, miR-101b, miR-145, miR-516-1, miR-527, miR-625†, miR-224, miR-290, miR-465, miR-201, miR-382, miR-505	82	53

† Indicate that type of miRNA arises from the 3′ arm of the mature sequence.

In addition, miRNA reverse transcription control and positive PCR control will be added for assessment of reverse transcription performance and PCR performance, respectively.

Development of the brain cancer miRNA profiling panel

Merging all collected data of 494 miRNAs from 4 commercial panels and 6 databases and meta-analysis is demonstrated in Supplementary Table 15. After intersection and filtration, the highly impacted 84 miRNAs were selected. Plate layout and assay location are depicted in Figure 5.

Functional enrichment analysis of the selected panel

The heat map of enriched KEGG pathway analysis demonstrated classification of the miRNAs into four distinct clusters (Figure 6A). Our miRNAs panel targeted 13 specific cancer pathways (Figure 6B), in addition to 78 cancer-related signaling pathways, including miRNAs in cancer, proteoglycans in cancer, pathways in cancer, ECM–receptor interaction, fatty acid biosynthesis, elongation, metabolism, and degradation, Hippo signaling pathway, TGF-β signaling pathway, p53 signaling pathway, cell cycle, adherens junction, and signaling pathways regulating pluripotency of stem cells (Figure 6C & Supplementary Table 16). Each cluster demonstrated on the heat map underwent GO analysis. Insignificant terms were filtered out to have 166, 187, 189 and 157 biological processes for green, orange, blue and red clades, respectively.

Venn diagram shows the common and unique GO terms for each clade (Figure 6D). A total of one quarter of the GO (56 terms, 24%) was unique for each clade. Interestingly, miRNAs clustered in the green clade (miR-122-5p,

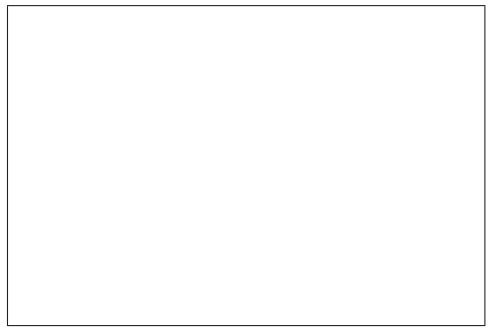

Figure 5. Plate layout of the putative custom miRNA panel for glioblastoma disease.

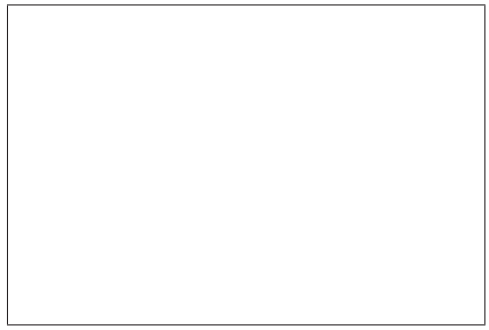

Figure 6. Functional enrichment analysis of the custom miRNA panel. (A) Heat map demonstrated hierarchical clustering of miRNAs into four distinct groups based on their enriched Kyoto encyclopedia of genes and genomes pathway analysis. Names of miRNAs in each cluster are shown at the right panel and colored according to their corresponding clade at the left panel. Glioma pathway is marked inside a red box on the bottom panel. Fisher's Exact Test (hypergeometric distribution) was adjusted as the enrichment analysis method and p-value threshold was set below 0.05. Color key at the top panel marks the p-value range. **(B)** List of predicted specific cancer pathways perturbed by the selected miRNAs in the custom panel. The number of enriched miRNAs and targeted genes in the pathway is notified on the bar of each KEGG pathway.**(C)** List of predicted signaling pathways to miRNAs in the custom panel. The number of enriched miRNAs and the number of targeted genes in the pathway are notified on the bar of each KEGG pathway.**(D)** Comparing gene ontology terms of four clustered miRNAs exhibited in hierarchical clustering analysis. Venn diagram shows the common and unique features of each clade (Data source: DIANA tool).

miR-126-5p, miR-129-5p, miR-130a-3p, miR-130b-3p, miR-132-3p, miR-140-5p, miR-148a-3p, miR-148b-3p, miR-181a-5p, miR-181b-5p, miR-181c-5p, miR-183-5p, miR-196b-5p, miR-19a-3p, miR-19b-3p, miR-218-5p, miR-27a-3p, miR-27b-3p, miR-30a-5p, miR-340-5p, miR-378a-3p and miR-485-5p) were highly enriched in catalytic activity and cell proliferation. Orange clade miRNAs (let-7a-5p, let-7b-5p, let-7d-5p, let-7e-5p, let-7f-5p, let-7g-5p, let-7i-5p, miR-106a-5p, miR-17-5p, miR-20a-5p, miR-424-5p, miR-497-5p, miR-7-5p, miR-93-5p and miR-95-5p) were involved in various cellular metabolism, chromatin remodeling with transcriptional activity and negative regulation of cell cycle. On the other hand, blue clade miRNAs (miR-103-53, miR-107, miR-142-5p, miR-143-3p, miR-15a-5p, miR-15b-5p, miR-16-5p, miR-18a-5p, miR-193b-3p, miR-195-5p, miR-21-3p, miR-221-3p, miR-222-3p, miR-22-3p, miR-224-5p, miR-23b-3p, miR-25-3p, miR-29a-3p, miR-29b-3p, miR-29c-3p, miR-34a-5p and miR-92a-3p) were shown to be impacted in chromatin remodeling with repression and regulation of apoptosis. Red clade miRNAs (miR-100-5p, miR-10a-5p, miR-10b-5p, miR-124-3p, miR-139-5p, miR-146-5p, miR-155-5p, miR-205-5p, miR-206, miR-21-5p, miR-26b-5p, miR-27a-5p, miR-31-5p, miR-425-5p and miR-99-5p) were significantly involved in the DNA repair system (nucleotide-excision repair, double-strand break repair and recombination) (Supplementary Table 17).

Discussion

The ongoing discovery of novel miRNAs in human genome and the lack of uniformity of the glioma disease-related miRNAs in commercial panels necessitate the updating and developing of a new comprehensive array with high diagnostic accuracy for GB clinical assessment.

By screening the current commercially available array platforms in the market, *in silico* database analysis and appraisal of the previous literature, we have developed a panel of highly enriched GB-specific miRNAs and that could have a potential diagnostic and/or prognostic role in GB patients. Most of these enriched miRNAs were reviewed in detail previously by Karsy *et al.* [29]. They have been implicated in diverse biological processes [30,31]. The oncomiR-21 was one of substantial miRNAs implicated in GB pathogenesis. It has been reported to downregulate many tumor suppressor and apoptotic genes including the mTOR pathway regulator *PTEN* (Phosphatase and tensin homolog), *PDCD4* (programmed cell death 4), *MTAP* (S-methyl-5′thioadenosine phosphorylase), *IGFBP3*, *RECK* (reversion inducing cysteine rich protein with Kazal motifs) and *TIPM3* (metallopeptidase inhibitor 3) [32–34]. Another putative oncomiR is miR-196a/b, which has been identified as one of a set of 35 miRNAs that are most frequently deregulated in glioma patients worldwide as highlighted in the meta-analysis of Piwecka *et al.* [35]. Guan *et al.* [36] also demonstrated that miR-196a/b upregulation was correlated with poorer survival and was one of the 16 deregulated miRs in GB in their miR microarray analysis. Regarding miR-221/222 paralogues, Zhang *et al.* [37] showed their enhancement of the GB malignant phenotype via activation of the p-Akt (phosphoprotein kinase B) pathway mediated by regulation of common gene expression. Additionally, the presence of a binding site for the miR-221/222 in the 3′ UTR of the protein tyrosine phosphatase μ, could explain in part the implication of miR-122/222 in GB tumorigenesis and invasiveness [38]. Furthermore, Quintavalle *et al.* [39] reported the association of miR-221/222 overexpression in GB with downregulation of the intracellular level of the alkylating enzyme O(6)-methylguanine-DNA methyltransferase which renders cells unable to repair genetic damage, contributing to increased markers of DNA damage and poor prognosis. The aforementioned miRNAs were included with others in the currently designed platform. They are involved in cellular metabolism, cell-cycle regulation, chromatin remodeling, cell proliferation, apoptosis and DNA repair system.

Divergent results of the available ready-made platforms and the previous studies may be related to the heterogeneous nature of the tumor samples themselves used in the studies (cultured cell lines or tumor tissues), the methodological approaches and the type of RNAs, which were applied for measurements standardization (e.g., let 7a, RNU6B or RNU24) [26]. We confirm that the current results concluded from the first phase of our project will warrant further validation of the developed miRNAs panel as a measure of tumor specificity and progression in the patient cohorts. This will be the next step in our ongoing project.

Summary points

- Ever since the discovery of miRNAs, scientists' perspective on cancer pathogenesis has completely overturned.
- Despite being a short sequence of nucleotides, miRNAs astonishingly have the ability to choreograph the expression of hundreds of other genes, making them capable of controlling tumor initiation, progression, invasion, metastasis and even response to treatment.
- Given its unique expression signature, proper miRNA profiling can help discriminate between glioblastoma (GB) and other types of brain tumors.
- The work presented herein represents a comprehensive data analysis of bioinformatics databases, previous literature and commercially available predesigned miRNA PCR arrays within the market, aiming at developing a more GB-specific and applicable custom-designed qRT-PCR miRNA assay.
- A highly enriched panel of 84 deregulated and GB-specific miRNAs has been developed.
- Application of this newly designed array especially in geographic regions with a limited financial support, after its validation, will save time and efforts of future researchers and can provide more accurate data about diagnosis and/or prognosis, as well as more personalized treatment options for GB patients.

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Authors' contributions

EA Toraih and HY Abdallah designed the work, EA Toraih contributed to the bioinformatics analyses. MS Fawzy critically revised the manuscript. All authors contributed in writing, reading and approval of the final manuscript.

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